MIL-S-14195A(OS)
17 February 1970
SUPERSEDING
MIL-S-14195(Ord)
22 November 1955

MILITARY SPECIFICATION

STYRENE MONOMER

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy.

1. SCOPE

1.1 This specification covers one grade of high purity styrene monomer (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-D-729	Drum: Metal, 55-Gallon (For Shipment of Non-corrosive Material)

STANDARDS

Federal

Fed. Test Method Std. No. 141	Paint, Varnish, Lacquer and Related Materials; Methods of Inspection, Sampling, and Testing

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking For Shipment and Storage

FSC 9330

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(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications - The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials

ASTM D 445	Viscosity of Transparent and Opaque Liquids (Kinematic and Dynamic Viscosities)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

3. REQUIREMENTS

3.1 Material - The styrene monomer shall be a clear colorless liquid aromatic hydrocarbon suitably inhibited to prevent premature polymerization.

3.2 Physical and chemical properties - When tested as specified in Section 4, the physical and chemical properties of the styrene monomer shall conform to the requirements specified in Table I.

TABLE I

PHYSICAL AND CHEMICAL PROPERTIES

PROPERTY	REQUIREMENTS	
	MINIMUM	MAXIMUM
Specific gravity (absolute) 25 degrees/25 degrees Centigrade (° C)	0.9035	0.9050
Refractive index, ND at 25° C	1.5434	-

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TABLE I (Continued)

PROPERTY	REQUIREMENTS	
	MINIMUM	MAXIMUM
Color	-	15 platinum-cobalt
Viscosity, centipoises at 25° C	-	0.80
Peroxides, as hydrogen peroxide by weight, percent	-	0.01
Polymer content, by weight, percent	-	0.20
Inhibitor, as para-tertiary (p-tert) butylcatechol, by weight, parts per million (ppm) (see 6.3)	10	50

3.3 Workmanship - The styrene monomer shall be processed in a manner that will produce the high quality product necessary to meet the requirements of this specification, and shall be free of defects which could cause the material to be unsuitable for the purpose intended.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection - The examination and testing of the styrene monomer shall be entirely by quality conformance inspection.

4.3 Test conditions - The laboratory testing conditions shall be in accordance with Fed. Test Method Std. No. 141 and as described herein.

4.4 Sampling -

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4.4.1 Quality conformance samples - Quality conformance samples shall consist of a sample for tests (4.4.1.2) and a sample for examination of filled containers (4.4.1.3). Samples shall be labeled completely with information identifying the purpose of the sample, name of product, specification number, lot and batch number, date of sampling, and contract number.

4.4.1.1 Inspection lot - An inspection lot size shall consist of the styrene monomer manufactured from one batch or blending operation, under essentially the same conditions and presented for acceptance at one time.

4.4.1.2 Sampling for tests - An 8-ounce sample shall be selected at random from each lot. The sample shall be placed in an airtight glass bottle having a glass stopper or cap containing a cork or cardboard liner covered with metal foil and labeled as specified in 4.4.1. All quality conformance tests shall be made on this sample.

4.4.1.3 Sampling for examination of filled containers - A random sample of one container shall be selected from filled containers fully prepared for delivery, in accordance with MIL-STD-105 at Inspection Level I and Acceptable Quality Level of 2.5 percent defective to verify all requirements of this specification in regard to packaging, marking, workmanship, and other requirements not involving tests.

4.4.1.4 Quality conformance inspection - The quality conformance inspection shall consist of the testing of the sample (4.4.1.2) as specified in 4.5, and an examination of the filled container (4.4.1.3) as specified in 4.6.

4.5 Test methods -

4.5.1 Specific gravity - Two specimens shall be tested in accordance with Fed. Test Method Std. No. 141, Method 4183, using a balance apparatus, except that the temperature shall be 25°/25° C, throughout. The value so obtained shall be converted to the approximate absolute specific gravity (in vacuum) by adding the factor 0.0002 to the apparent specific gravity in air.

4.5.2 Refractive index - The refractive index of 2 specimens shall be determined with an Abbe refractometer or any equally accurate instrument at 25° ±1° C to an accuracy of ±0.0001 unit, and using light having a wave length equivalent to the "D" line of sodium.

4.5.3 Color -

4.5.3.1 Preparation of color standards - Color standards shall be prepared as follows:

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Dissolve 1.245 grams (gm.) of potassium chloroplatinate (K_2PtCl_6) containing 0.500 gm. of platinum, and 1.000 gm. of crystallized cobaltous chloride ($CoCl_2.6H_2O$), containing about 0.278 gm. of cobalt, in water containing 100 milliliters (ml.) of concentrated hydrochloric acid, and dilute to 1 liter with distilled water. If K_2PtCl_6 is not available, the following procedure shall be used:

Dissolve 0.5 gm. of metallic platinum in aqua regia. Remove the nitric acid by repeated evaporation to dryness after adding an excess of hydrochloric acid. Dissolve this product with 1.000 gm. of $CoCl_2.6H_2O$.

4.5.3.2 Procedure - Fifty ml. of the sample shall be transferred to a 50-ml. Nessler tube. The color of the sample shall be compared with color standards prepared by diluting 0.5, 1.0, 1.5, and 2.0 ml. of the standard solution (see 4.5.3.1) to 50 ml. in similar 50-ml. Nessler tubes. The platinum cobalt number of the sample shall be the number of milligrams of platinum contained in 1000 ml. of the standard solution which matches the sample. The standard solution has a color of 500. The Nessler tube containing 1.5 ml. of the standard solution diluted to 50 ml. has a color of 15 platinum-cobalt (see 6.4).

4.5.4 Viscosity - Two specimens shall be tested, using a modified Ostwald viscosimeter, in accordance with ASTM Method D 445, except that the temperature of the water bath may be allowed to fluctuate plus or minus 0.1° C and the bath temperature need be read only to the nearest 0.1° C. The value reported shall be absolute viscosity in centipoises instead of kinematic viscosity in centistokes. The viscosity shall be calculated as follows:

$$\text{Viscosity at 25° C, centipoises} = \frac{t_1 \times d_1 \times n}{t_2 \times d_2}$$

where:

t_1 = time of flow of the unknown at 25° C.
t_2 = time of flow of the standard material (Bureau of Standards or API oil) at 25° C.
d_1 = density of the unknown, grams per milliliter (gm./ml.) at 25° C. (d_1 may be determined by adding 0.0027 to the value of specific gravity 25°/25° C determined as specified in 4.5.1).
d_2 = density of standard material gm./ml. at 25° C.
n = viscosity, centipoises, of the standard material at 25° C.

4.5.5 Peroxides -

4.5.5.1 Reagents - Approximately 0.02 normal (N) solution of titanous chloride shall be made by diluting the 20 percent reagent grade solution. Ferric ammonium sulfate solution shall be made by dissolving 35.1 gm. of crystallized ferrous ammonium sulfate hexahydrate in 400 ml. of distilled water. Twenty-five ml. of concentrated sulfuric acid shall be added, and the solution warmed slightly. Potassium permanganate

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solution shall now be added until the iron is completely oxidized, and the solution is diluted to 1 liter with distilled water. One ml. of this solution will contain 0.005 gram of ferric iron. Ferrous ammonium sulfate solution shall be made by dissolving 19.6 gm. of hexahydrate salt in 800 ml. of distilled water and adding 50 ml. of concentrated sulfuric acid. After solution is complete, dilution should be made to 1 liter. The ferrous iron solution shall be approximately 0.1N and does not require standardization. A 10-percent aqueous solution of potassium thiocyanate shall be used as an indicator; sulfuric acid diluted to a ratio of 1:1 with distilled water, concentrated hydrochloride acid, and absolute methanol are also necessary for carrying out the determination.

4.5.5.2 Apparatus - A 50-ml. long-tipped burette shall be used for the titanous chloride solution. If the determination is to be run each day, it is advantageous to assemble equipment for storing the titanous chloride solution under an inert atmosphere (see 6.5). The nitrogen used shall be passed through a pyrogallic acid solution to remove the oxygen and a slow flow of nitrogen shall be maintained at all times. A satisfactory apparatus is shown in Figure 1.

4.5.5.3 Standardization of titanous chloride solution - Ten ml. of the standard ferric ammonium sulfate solution shall be pipetted into a 250-ml. wide-mouth Erlenmeyer flask containing 50 ml. of distilled water, 15 ml. of concentrated hydrochloric acid, and 10 ml. of the potassium thiocyanate indicator. A small lump of "dry ice" shall be added to the flask in order to maintain an inert atmosphere over the solution at all times. The tip of the burette shall be kept low in the flask and the titration with titanous chloride solution shall be carried out until the pink color of the ferric thiocyanate has just disappeared. This titration shall be recorded.

If the peroxide is to be considered as diethyl peroxide, then the following calculation shall be made to obtain the factor of the titanous chloride solution:

$$\text{factor} = \frac{0.05 \times 45.04}{Ts \times 55.85}$$

where:

Ts = ml. of titanous chloride required for titration of the 0.05 gm. of ferric iron contained in 10 ml. of standard iron solution.

factor = gm. of diethyl peroxide equivalent to 1 ml. of the titanous chloride.

4.5.5.4 Procedure - Five ml. of the potassium thiocyanate solution, 2 ml. of the dilute sulfuric acid, and 10 ml. of the approximately 0.1N ferrous ammonium sulfate solution shall be pipetted into a 250-ml. wide-mouth Erlenmeyer flask containing 100 ml. of the methanol. A small lump of "dry ice" shall be placed in the flask, the solution mixed by swirling, the flask covered with a watch glass, and allowed to stand for 15 minutes. While the burette is low in the flask, the titanous chloride solution shall be carefully added until the pink color just disappears. This amount of titanous chloride is not required in the calculation since this procedure

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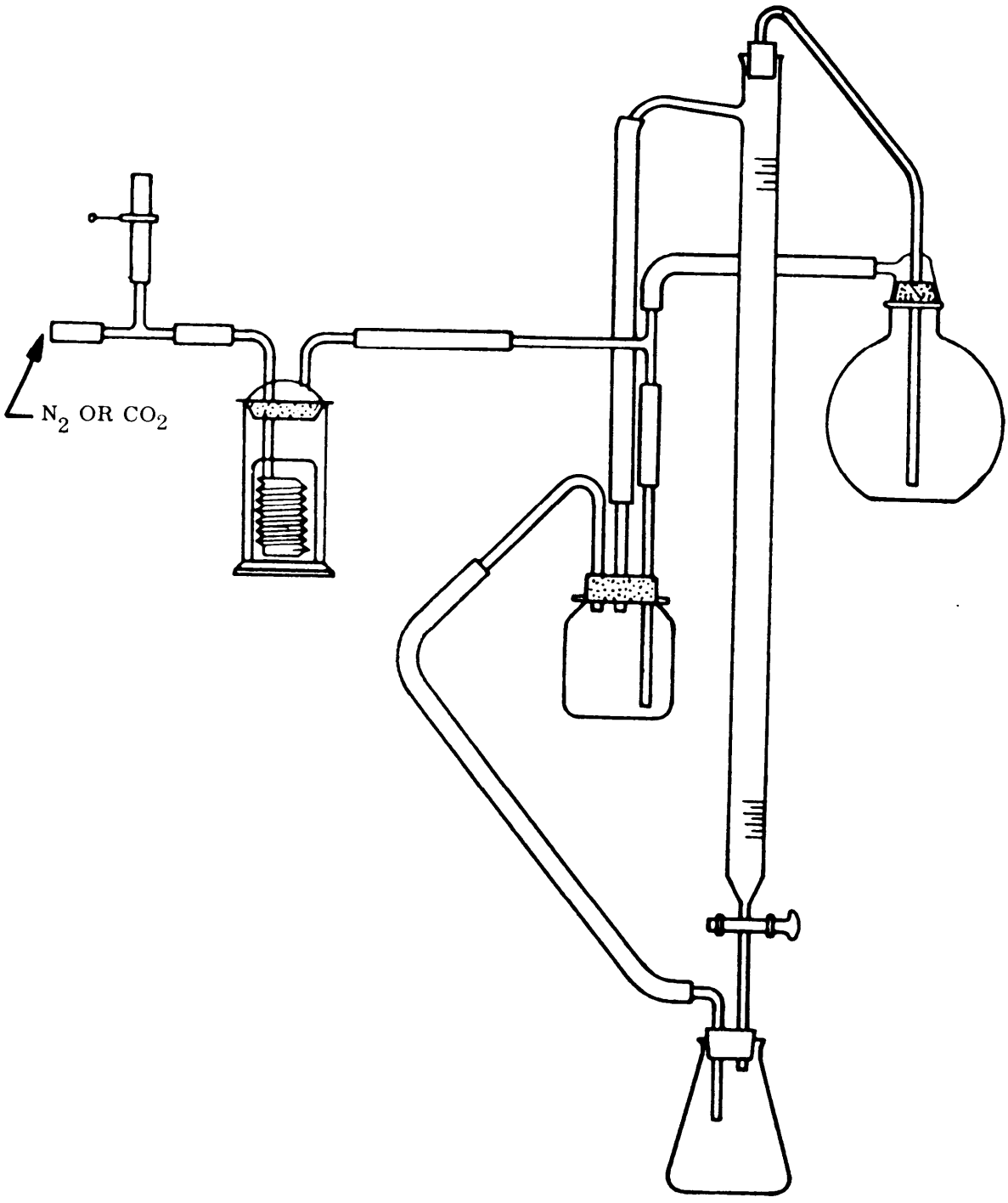


Figure 1. Apparatus for titrations with titanous chloride

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is intended only to reduce any ferric sulfate present in the ferrous ammonium sulfate solution used. The measured sample of styrene shall then be added. Twenty-five ml. of styrene monomer shall be pipetted, mixed, covered with a watch glass and allowed to stand for 2 hours. The ferric sulfate formed from the reaction shall then be titrated with a standard titanous chloride solution until the pink color of the ferric cyanate just disappears. The titration shall be carried out in a manner identical with the one used to dispel the ferric sulfate from the ferrous ammonium sulfate. The weight percent of diethyl peroxide shall be calculated as follows:

$$\text{Percent by weight of peroxides as diethyl peroxide} = \frac{T \times \text{factor} \times 100}{25 \times 0.9}$$

where:

T = ml. of standard titanous chloride required for the ferric iron produced by the peroxides in the sample.

factor = gm. of diethyl peroxide equivalent to 1 ml. of the titanous chloride as determined in the standardization

4.5.6 Polymer content -

4.5.6.1 Quick alcohol method - Where the amount of polymer is negligible and rapid work is desired, the following method shall be used: Determine the amount of polymer at the time of assay by placing 10 ml. of dry methanol in a test tube and adding 1 to 2 ml. of styrene. If, after shaking the mixture vigorously for a few seconds, only a light cloud of precipitate is produced, the material shall be considered satisfactory. However, if a heavy precipitate is formed, use the gravimetric method (see 4.5.6.2).

4.5.6.2 Gravimetric method - The polymer content shall be determined as follows: Add 25 gm. of the sample to 100 ml. dry chemically pure (c.p.) methanol in a 250-ml. beaker. Stir this mixture well and place on a steam bath for 5 minutes or longer to aid coagulation. Allow the solution to cool and filter on a tared Gooch crucible or a fine porosity fritted glass crucible. Place the precipitate under an infrared lamp for 5 minutes and then dry to a constant weight at 70° C. Cool the polymer in a desiccator and weigh. Calculate the polymer content as follows:

$$\text{Percent by weight of polymer} = \frac{\text{gm. polymer} \times 100}{\text{gm. sample}}$$

4.5.7 Inhibitor - Presence, ppm, of inhibitor as TBC shall be determined by Method A (4.5.7.1) or Method B (4.5.7.2).

4.5.7.1 Method A -

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4.5.7.1.1 Reagents - Sodium hydroxide solution, 1N is needed for this determination. Styrene stock solution, containing 1gm. of p-tert butylcatechol (TBC) per liter, shall be prepared. If inhibitor-free monomer is not available, it may be obtained by washing the inhibited material with dilute caustic solution and drying over anhydrous calcium sulfate.

4.5.7.1.2 Procedure - Ten ml. of styrene and 5 ml. of the 1N sodium hydroxide solution shall be placed in a small separatory funnel and shaken for 3 to 5 minutes. The solution shall be allowed to settle and the aqueous layer then drawn off into a 5-ml. Nessler tube. Using the styrene monomer stock solution, knowns shall be made up in the range of TBC expected and handled in the same manner as the sample. The samples shall be matched with the knowns to find the ppm of inhibitor in the sample.

4.5.7.2 Method B -

4.5.7.2.1 Equipment - The equipment required for the performance of Method B test for the presence of inhibitor shall be as follows:

(a) Bausch and Lomb "Spectronic 20" spectrophotometer with blue sensitive phototube, and 11 mm path cuvette.

(b) Beakers: two-150 ml., one-600 ml., one-1000 ml.

(c) Crucible: Sintered glass filtering, high form crucibles, coarse and fine porosity.

(d) Graduated cylinder: one-50 ml.

(e) pH meter: Beckman "Zeromatic" or equivalent

(f) Pipets: one-5 ml., one-5 ml. graduated, two-2 ml., one-10 ml., one-50 ml.

(g) Separatory funnels: one-125 ml. with Teflon plug

(h) Volumetric flasks: seven-50 ml., two-100 ml., one-1000 ml.

4.5.7.2.2 Reagents - Reagents required for the performance of Method B tests for the presence of inhibitor shall be prepared as follows:

(a) pH 3.5 buffer solution - Fifty ml. of reagent grade acetic acid shall be added to 1 liter of water. With the pH meter, the pH of the solution shall be adjusted to 3.5, using reagent grade sodium acetate.

(b) Ferric chloride solution - One-half gm. of reagent grade $FeCl_3.6H_2O$ shall be dissolved in 100 ml. of distilled water (the $FeCl_3.6H_2O$ shall be stored in a desiccator).

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(c) o-phenanthroline solution - One-half gm. of o-phenanthroline monohydrate (G. Frederick Smith Co. or equivalent) shall be dissolved in 100 ml. of distilled water. The flask may be warmed, if necessary, to effect complete dissolution.

(d) Recrystallized TBC - A preliminary recrystallization shall be made by dissolving 4 gm. of TBC in 5 gm. of carbon tetrachloride using heat. The solution shall be quickly cooled by placing the beaker in ice water, and shall be seeded with two or three crystals from the TBC bottle. The solution shall not be stirred until the crystals have formed and the solution is cold. The solution shall then be filtered slowly through a coarse porosity sintered glass crucible. The crystals shall then be dried for 15 to 20 minutes in an oven maintained at 30 to 35° C. To prepare the crystals of TBC for the stock TBC solution, 21 gm. of TBC shall be dissolved in 30 gm. of carbon tetrachloride using heat. Recrystallization shall be as above, using the crystals from the preliminary recrystallization for seeding. The recrystallized TBC shall have a minimum melting point of 49° C.

(e) Stock TBC solution - Standard solutions shall be made up using freshly recrystallized TBC. 0.1 gm. of TBC solution shall be weighed into a 100-ml. volumetric flask and made to 100 ml. with pH 3.5 buffer solution. A 1 ml. aliquot of this solution shall be diluted with buffer solution to 100 ml. in a volumetric flask. This final solution contains 10 micrograms of TBC per ml.

4.5.7.2.3 Preparation of standard TBC curve - The standard TBC curve shall be prepared in the following manner:

(a) Using a graduated pipet, 1, 2, 3, and 4 ml. respectively of the stock TBC solution containing 10 micrograms of TBC per ml. shall be placed in four 50-ml. volumetric flasks.

(b) Thirty ml. of pH 3.5 buffer solution shall be added to each of the four volumetric flasks, and also to a fifth, that is to be used as a reagent blank.

(c) 2 ml. of ferric chloride solution shall be pipetted into each volumetric flask, and the solution shall be swirled.

(d) 2 ml. of o-phenanthroline solution shall be pipetted into each volumetric flask, and the solution swirled; make each solution to 50 ml. with distilled water. The solution shall set for one hour.

(e) When 1/2 hour has elapsed, the "Spectronic 20" shall be turned on.

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(f) When 1 hour has elapsed, the "Spectronic 20" shall be set at 0 percent T with no cell in the instrument. It shall be set at 100 percent T with distilled water in the cell (an 11 mm cuvette).

(g) The absorbance of each of the solutions shall be measured at 500 millimicrons.

(h) The corrected absorbance of each standard solution shall be obtained by subtracting the absorbance of the reagent blank from each total absorbance.

(i) These corrected absorbances shall be plotted versus total micrograms of TBC in the final 50 ml. solution.

(j) If the standard curve does not go through the origin, the recrystallization of the TBC shall be repeated, and a new standard curve shall be prepared.

4.5.7.2.4 Procedure for determination of TBC in styrene - The following procedure shall be followed to determine TBC in styrene:

(a) Twenty gm. of styrene shall be weighed into a 125-ml. separatory funnel.

(b) With a pipet, 10 ml. of ferric chloride solution shall be added.

(c) The separatory funnel shall be stoppered and vigorously shaken for five minutes; the aqueous layer shall be drawn off into a 50-ml. volumetric flask.

(d) The funnel stem shall be washed down with 2 ml. of distilled water.

(e) Steps (b), (c), and (d) shall be repeated, and the washings shall be collected in a 50-ml. volumetric flask.

(f) 10 ml. of distilled water shall be added to the separatory funnel. The funnel shall be shaken for 3 minutes and the aqueous layer shall be drawn off into the volumetric flask.

(g) The solution in the volumetric flask shall be allowed to stand for 10 minutes; if a high degree of turbidity is still present at the end of this time, the solution shall be filtered through a fine crucible into a 50-ml. volumetric flask, washing the crucible with a small amount of buffer solution. (Filtration will probably not be necessary, since some turbidity is permissible.)

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(h) This solution shall be made to 50 ml. with buffer solution; stoppered and shaken.

(i) With a pipet, 5 ml. of this solution shall be added to a second 50-ml. volumetric flask containing 30 ml. of buffer solution.

(j) The reagent blank shall be prepared by pipetting 2 ml. of ferric chloride solution into a 50-ml. volumetric flask, and adding 30 ml. of buffer solution.

(k) 2 ml. of o-phenanthroline solution shall be pipetted into each volumetric flask, including the reagent blank, and the solution shall be swirled. Each solution shall be made to 50 ml. with distilled water, and shall be allowed to stand for 1 hour.

(l) When 1/2 hour has elapsed, the "Spectronic 20" shall be turned on.

(m) When 1 hour has elapsed, the "Spectronic 20" shall be set at 0 percent T with no cell in the instrument. It shall be set at 100 percent T with distilled water in the cell (an 11 mm cuvette).

(n) The absorbance of each of the solutions shall be measured at 500 millimicrons.

(o) The corrected absorbance of each standard solution shall be obtained by subtracting the absorbance of the reagent blank from each total absorbance.

(p) Using the corrected absorbance of the sample, the total micrograms of TBC in the final solution shall be read from the standard curve prepared as directed in 4.5.7.2.3. The following formula shall be used to calculate parts per million TBC in the sample:

$$\text{ppm TBC} = \frac{\text{micrograms from standard curve x 50 ml.}}{\text{sample weight x final aliquot taken}}$$

4.5.7.3 Referee test - In cases of dispute, the result obtained by Method B shall take preference.

4.6 The sample of 4.4.1.3 shall be inspected for conformance with the applicable sections of Section 5.

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4.7 Rejection criteria - When any lot quality conformance test sample fails to meet any of the test requirements of this specification or when the number of defective filled containers exceeds the acceptance number as specified in 4.4.1.3, the lot represented by the sample shall be rejected.

5. PREPARATION FOR DELIVERY

5.1 Levels of packaging - The level of packaging shall be Level A or C, as specified. (See 6.2.)

5.1.1 Level A - Unless otherwise specified by the procuring activity, the styrene monomer shall be packaged in 55-gallon drums in accordance with PPP-D-729, Type III.

5.1.2 Level C - When Level C is required, packaging shall conform to the manufacturer's commercial practice, unless the procuring activity determines that modification to the manufacturing commercial practice is required to meet the requirements of this level.

5.2 Packing - Not required.

5.3 Marking - Except as specified herein or otherwise specified by the procuring activity, marking and labeling of all containers shall comply with MIL-STD-129.

5.3.1 Identification information - Each container shall be marked on one face or side in black ink with the following information arranged as indicated:

Stock Number
Nomenclature: Styrene Monomer
Quantity
Specification MIL-S-14195A(OS)
Lot number
Date of Manufacture
Manufacturer's Name or Trade Mark
Contractor (If other than the Manufacturer).
Contract or Order Number
Preferably use within days following date of manufacture

5.3.2 Instructions for storage and issue - The following information and instructions shall appear on each unit package:

Storage instructions - (The recommendations of storage temperature and periods to prevent premature polymerization as established by the manufacturer). "Stocks of this material shall be kept at an operating minimum and the

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styrene monomer should be issued from stock on a basis of "first-in, first-out" so that no monomer is rendered useless by deterioration resulting from excessive time in storage."

6. NOTES

6.1 Intended use - The styrene monomer is the standard commercial material used in all general purpose applications and is sufficiently pure to be used for applications such as optical cements which require high purity. Its chief applications are expected to be for the preparation of cements for joining various materials and as a component for plastic bonded explosives.

6.2 Ordering data - Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Selection of applicable levels of packaging (see 5.1).

6.3 It is recommended that quantities of inhibitor in the lower range be used in material expected to be delivered in cool weather and quantities of inhibitor in the higher range be used in material expected to be delivered in warm weather.

6.4 A satsifactory method of making color comparison is described in "Standard Methods of Water and Sewage," 8th Edition, part 1, section VIA - Color-Comparison with Platinum-Cobalt Standards, prepared and published by American Public Health Association.

6.5 A satisfactory method for storing and using the titanous chloride solution, prepared as specified in 4.5.5, under an inert atmosphere is described in "Scotts' Standard Methods of Chemical Analysis," 5th Edition, Volume 1, page 479.

Preparing activity:
Navy - OS
(Project No. 9330-N405)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-S-14195A	

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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